POLAMREDDY REVANTH REDDY

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SUMMARY

Computer Science undergraduate specializing in AI development, with research internship experience at IIT Tirupati and IIT Madras. Experienced in building and shipping end-to-end AI applications — from model training through backend integration to deployed, user-facing products — across computer vision, classification, and reinforcement learning agents. Skilled in Generative AI, LLMs, and RAG, with full-stack ability (Flask/FastAPI, Node.js, React.js) to productionize AI systems.

EXPERIENCE

Full Stack Developer – Enterprenex Solutions Pvt Ltd **Aug 2026 – Present**
IT Department (Remote/Online) *Waluj, Chhatrapati Sambhajinagar, Maharashtra*

- Contributing to the end-to-end development of **ENX Money**, a business finance and expense management platform integrating accounting, GST invoicing, inventory, and ledger management features.
- Developing and integrating **frontend interfaces and RESTful APIs** for authentication, business profiles, customers, suppliers, khata ledger, products, inventory, sales invoices, purchase bills, payments, and financial transactions.
- Working with **Node.js, Express.js, MySQL, JWT, and Git/GitHub** to build backend services, implement database operations, handle authentication, validate requests, and maintain secure data handling.
- Integrating frontend components with backend APIs, coordinating application state and data flow, and collaborating with backend and data analytics teams to support dashboard functionality and reliable application performance.
- Implementing and debugging end-to-end application features, testing API integrations, resolving frontend-backend issues, and improving code quality, maintainability, and overall user experience.
- Following a structured **Git-based development workflow**, including feature branches, code integration, testing, documentation, and deployment preparation.

Research Intern – IIT Tirupati, SEVA Lab **Jan 2026 – Jun 2026**
Under Dr. Chalavadi Vishnu *Remote*

- Contributed to an **Edge-Semantic Collaborative Network for Camouflaged Object Detection**, focusing on deep learning-based feature learning and model development for autonomous systems.
- Applied computer vision techniques and deep learning frameworks to develop, optimize, and evaluate research-oriented model pipelines.

PROJECTS

EcoSort AI System — *Python, PyTorch, Flask, React.js, SQLite* Live Demo

- Built a waste classification system using MobileNetV2 transfer learning, served via a Flask REST API with a React.js frontend for real-time prediction.
- Integrated an analytics dashboard and SQLite database; deployed as a production-ready full-stack AI application.

Flappy Bird Reinforcement Learning Agent — *Python, DQN, PyGame, Streamlit* Live Demo

- Developed an autonomous DQN-based agent achieving high obstacle-avoidance performance; improved training stability using experience replay and target networks.

Crop Disease Advisory System — *Python, TensorFlow, CNN, Flask* Live Demo

- Built a CNN-based plant disease classifier with an end-to-end pipeline (preprocessing, training, evaluation, real-time inference), deployed as a web app returning predictions and treatment recommendations.

Image Search Engine — *Python, Flask, Deep Learning* Live Demo

- Built a deep-learning embedding-based image retrieval system with a Flask REST API for feature extraction and similarity search.

Sepsis Detection Using Machine Learning — *Python, Scikit-learn, Flask* GitHub

- Built an early sepsis prediction model on ICU data using SMOTE for class imbalance; evaluated with ROC-AUC and deployed via a Flask prediction API.

SKILLS

AI Development: Model Training & Evaluation, Transfer Learning, Model Serving, Generative AI, LLM, RAG, Reinforcement Learning, Computer Vision
Frameworks/Libraries: PyTorch, TensorFlow, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, Seaborn
Full-Stack & Deployment: Flask, FastAPI, Node.js, Express.js, React.js, REST APIs, Streamlit, Render, Netlify
Languages: Python, C/C++, R, SQL **Databases:** SQLite, MySQL, MongoDB **Cloud & Tools:** AWS, IBM Cloud, GitHub Actions, Git, Jupyter

EDUCATION

Chhattisgarh Swami Vivekanand Technical University, Bhilai **2023 – 2027**
B.Tech (Honours) in CSE (Data Science) CGPA: 7.8 / 10 (till 6th Semester)
Narayana Junior College (Class XII: 89.6%), 2021 – 2023 | Narayana School (Class X: 99.5%), 2020 – 2021

CERTIFICATIONS

Complete Python for Data Science and Machine Learning – Krish Naik | AWS Training Certificate – Amazon CodeWhisperer
Data Science Workshop – Mindenious | Data Analysis using Excel – Great Learning